

Manufacturer Design & Simulation Tools

A reference list of free, vendor-hosted design/simulation tools — the WEBENCH-style "enter your specs, get a circuit" tools, plus configurators, selectors, and calculators from silicon and component manufacturers. Compiled September 2026.

1. Power Electronics / Power Supply Design

Tool	Vendor	What it does
WEBENCH Power Designer / Circuit Designer	Texas Instruments	The original: enter Vin/Vout/Iout and get a ranked list of complete power-supply designs (schematic, BOM, simulation, CAD export). Still actively maintained.
ADI Power Studio	Analog Devices	Umbrella suite launched Oct 2025: Power Studio Planner (system-level power-tree architecture) + Power Studio Designer (IC-level design with live efficiency/loss analysis). Also houses the older LTpowerCAD®/LTpowerPlanner® and LTspice®.
PowerEsim	Infineon	Full SMPS + transformer design tool — loss analysis, thermal simulation, many topologies, not Infineon-only.
Elite Power Simulator	onsemi	Cloud simulator for EliteSiC MOSFETs/IGBTs/IPMs — device-level switching behavior and PLECS system models, aimed at higher-power SiC designs.
WebDesigner+ Power Supply	onsemi	Enter I/O requirements, get recommended PMICs, topology, and passives — closest onsemi equivalent to WEBENCH's original flow.
DC/DC Designer Online	Monolithic Power Systems (MPS)	Pick an MPS regulator + target specs → simulated circuit, schematic, layout guide, BOM.
eDesignSuite – Power Management Design Center	STMicroelectronics	No-login suite: DC/DC & AC/DC converter design, multi-rail Power Tree Designer , and a Digital Power Workbench for STM32-based digital control loops.

Tool	Vendor	What it does
Power System Designer (PowerBench)	Vicor	Architect multi-output power chains from Vicor's brick/module portfolio; up to 75% faster than manual selection per Vicor.
REDEXPERT	Würth Elektronik	Component search <i>and</i> design tool for inductors/ferrites/capacitors. The DC-DC Converter Designer module simulates real (not datasheet-ideal) inductor/MLCC behavior and recommends a controller IC — very useful alongside any of the above.

2. Analog Electronics / Signal Chain

Tool	Vendor	What it does
Analog Filter Wizard	Analog Devices	Design active low/high/band-pass filters with real op-amps; shows ideal vs. real-world response, exports SPICE files.
ADIsimRF	Analog Devices	RF signal-chain calculator — cascaded gain, noise figure, IP3, P1dB, power consumption across up to 20 stages.
ADIsimPLL	Analog Devices	PLL synthesizer design/simulation — phase noise, fractional-N spurs, anti-backlash pulse. (Same design-tools hub as above.)
DiffAmpCalc / Precision ADC Driver Tool / Photodiode Wizard / Virtual Eval	Analog Devices	Suite of interactive web tools for signal-conditioning circuits — differential amps, ADC drivers, photodiode transimpedance stages, and ADC/DAC performance simulation.
LTspice	Analog Devices	ADI's preferred SPICE simulator (ADIsimPE was sunsetted Sept 2019); free, huge model library.
ANALOG-FILTER-DESIGNER	Texas Instruments	Downloadable filter design tool (TI's counterpart to ADI's Filter Wizard).

3. Embedded Systems — MCU Configuration & Code Generation

Tool	Vendor	What it does
STM32CubeMX	STMicroelectronics	Graphical pin-mux, clock-tree, and peripheral configurator for STM32 MCUs/MPUs; auto-generates HAL/LL init C code and detects conflicts. Standalone or built into STM32CubeIDE.
SysConfig	Texas Instruments	Graphical pin-mux and driver/stack configurator across MSPM0, C2000, SimpleLink, and MSP430 families; generates commented C code, real-time conflict checking.
MPLAB Code Configurator (MCC)	Microchip	Peripheral/driver code generator plugin for MPLAB X — covers 8/16/32-bit PIC (including PIC32/Harmony), AVR, and dsPIC. Two flavors: MCC Classic and the newer MCC Melody .
esp-gpio-tool	Espressif	Official YAML-based GPIO/pin-constraint validator for ESP32 designs (data also powers the community reference site esp32pin.com).

4. Digital Systems / FPGA

Tool	Vendor	What it does
Xilinx Power Estimator (XPE) / Power Design Manager	AMD	Spreadsheet/GUI tool for pre-design FPGA power budgeting (static + dynamic) across logic, I/O, and transceivers; PDM is the newer platform for Versal/Kria devices. Pairs with Vivado.
Early Power Estimator (EPE) / Power and Thermal Calculator (PTC)	Altera (Intel)	Same role for Altera FPGAs — early-stage power/thermal estimation before or during Quartus Prime implementation.

5. Clocks & Timing

Tool	Vendor	What it does
ClockBuilder Pro (CBPro)	Skyworks (formerly Silicon Labs' timing line)	Step-by-step clock-tree design for clock generators, jitter attenuators, and buffers; validates feasibility, recommends devices, generates register maps and production files.
Clock Tree Architect	Texas Instruments	Clock-tree synthesis tool that generates solutions from system requirements (listed under TI's design/simulation tools hub).

6. Communications Protocols / RF & Interfaces

Tool	Vendor	What it does
CAN Bus Bit Timing Calculator (+ CAN FD version)	Kvaser	Calculates valid CAN/CAN-FD bit-timing register sets (prescaler, segments, SJW, sample point) for a given clock and bus speed.
MatchCalc	Qorvo	RF/microwave impedance-matching calculator from S1P/S2P files, with live Smith Chart tuning.
RF Impedance Matching Calculator	Qorvo	Quick L-match network calculator (C and L values) for matching amplifier stages.

Notes

- Most of these require a free account only for saving/exporting designs — browsing and basic use is generally open.
- Several vendors merged or rebranded recently: Maxim → Analog Devices (shared toolset); Silicon Labs' timing business → Skyworks (ClockBuilder Pro).
- TI's WEBENCH Amplifier/Sensor Designer tools were discontinued in Jan 2020 (Adobe Flash EOL) — only the **Power Designer/Circuit Designer** side of WEBENCH survives, and it's still being updated.

- REDEXPERT and MatchCalc/RF tools pair well with the power/analog tools above — use them to select the actual magnetics or RF matching components once a controller IC is chosen.